

SUMMARY

Computer Science student (2023–2027) specializing in GenAI engineering and backend development. I build production-ready AI applications — RAG pipelines, LangGraph agentic systems, NL-to-SQL agents, and scalable Django backends. My focus is on bridging cutting-edge language models with real-world, deployable systems that solve actual problems.

EDUCATION

Bachelors in computer science and information technology

Orchid International College, Kathmandu · 2023 – 2027

Higher Secondary (+2)

Trinity International College, Kathmandu · 2020-2022

Secondary Education Examination (SEE)

Tamor Valley Secondary Boarding School, Dhankuta

SKILLS

AI / LLM	LangChain, LangGraph, Groq (LLaMA 3.3 70B), OpenAI SDK, RAG, Agentic Systems, Tool Calling, MCP
Backend	Python, FastAPI, Django, REST APIs, PostgreSQL, SQLite, ChromaDB (Vector DB)
Frontend	React (Vite), JavaScript, HTML/CSS, Tailwind CSS
DevOps	Git, GitHub, Render, Cloudflare, Docker (basic), uv package manager

PROJECTS

AstroQuery — RAG Conversational Interface for Space Data

- Built end-to-end Django RAG system for natural-language querying of satellite orbital CSV databases and mission PDF documents
- Stack: ChromaDB · Groq LLaMA 3.3 70B · OpenAI · LangChain · cosine similarity top-4 retrieval (chunk 1,000 / overlap 200)
- Implemented automated online evaluation metrics per query: retrieval relevance, faithfulness, and completeness

AI-Assistant Portfolio Website

- RAG chatbot (Groq · Django · ChromaDB) for intelligent Q&A over resume and project data
- Semantic vector retrieval with context-aware multi-turn conversation design
Deployed to Render; custom domain via Cloudflare DNS (bigyan-luitel.com.np)

NL-to-SQL LangGraph Agent

- LangGraph · Groq LLaMA 3.3 70B · SQLite · Python
- Designed and built a schema-aware NL-to-SQL agent with evaluator node, MemorySaver for multi-turn conversation history, and Groq LLaMA 3.3 70B as backbone LLM.
- Implemented LangGraph state machine with conditional routing between SQL generation, evaluation, and correction nodes.
- Extending with Plotly analyst node for data visualization, FastAPI HTTP endpoint, and PostgreSQL migration.

CERTIFICATE

AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents
Credential ID: UC-0c58df77-a37e-4b79-8a32-ce42087b6075
Credential URL: https://ude.my/UC-0c58df77-a37e-4b79-8a32-ce42087b6075